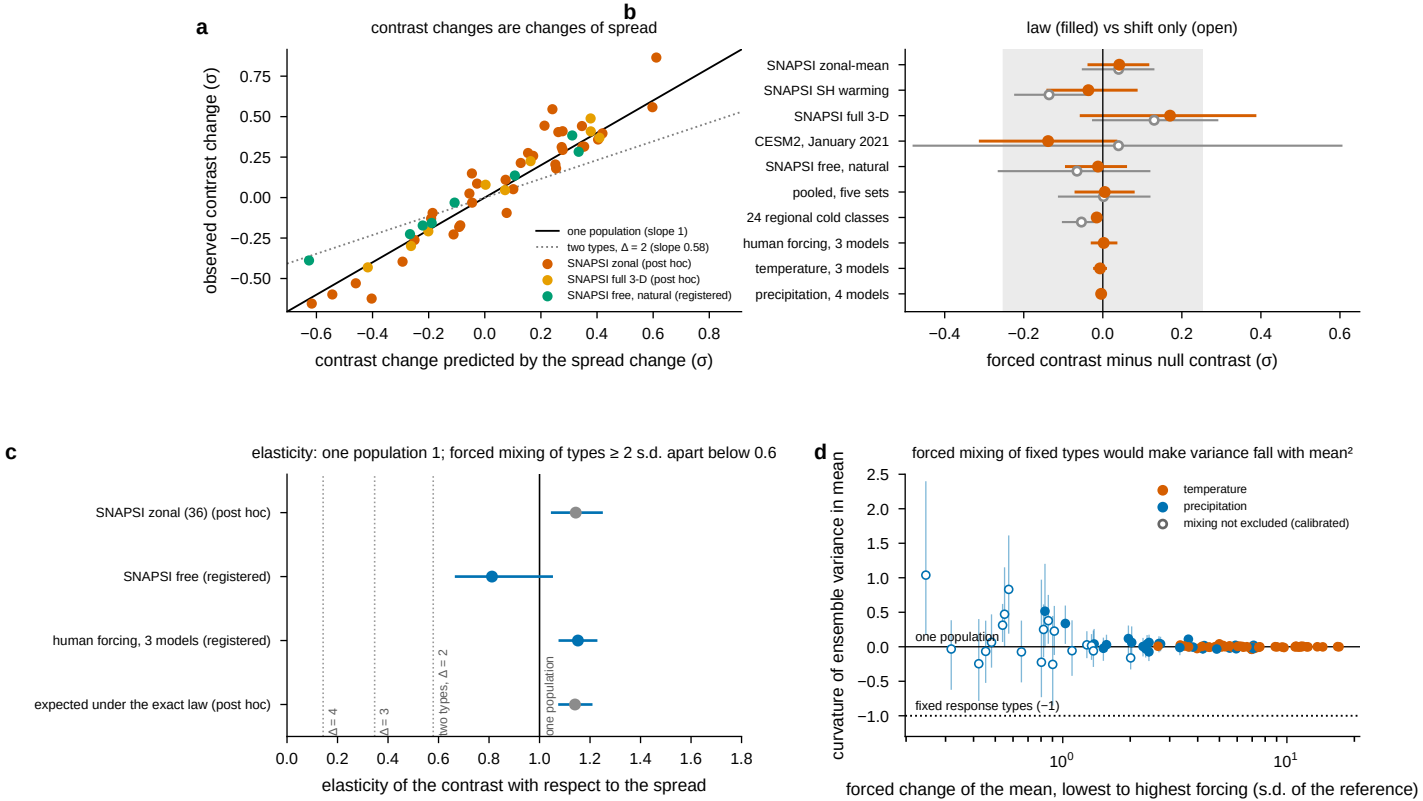
a
contrast changes are changes of spread
observed contrast change (σ)
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
contrast change predicted by the spread change (σ)
one population (slope 1)
two types, Δ = 2 (slope 0.58)
SNAPSI zonal (post hoc)
SNAPSI full 3-D (post hoc)
SNAPSI free, natural (registered)
b
law (filled) vs shift only (open)
SNAPSI zonal-mean
SNAPSI SH warming
SNAPSI full 3-D
CESM2, January 2021
SNAPSI free, natural
pooled, five sets
24 regional cold classes
human forcing, 3 models
temperature, 3 models
precipitation, 4 models
−0.4
−0.2
0.0
0.2
0.4
0.6
forced contrast minus null contrast (σ)
c
elasticity: one population 1; forced mixing of types ≥ 2 s.d. apart below 0.6
SNAPSI zonal (36) (post hoc)
SNAPSI free (registered)
human forcing, 3 models (registered)
expected under the exact law (post hoc)
Δ = 4
Δ = 3
two types, Δ = 2
one population
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
elasticity of the contrast with respect to the spread
d
forced mixing of fixed types would make variance fall with mean²
curvature of ensemble variance in mean
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
temperature
precipitation
mixing not excluded (calibrated)
one population
fixed response types (−1)
10^0
10^1
forced change of the mean, lowest to highest forcing (s.d. of the reference)